Kaiyao Duan

Sophia Antipolis, France | enjoy_sunshine@icloud.com | github.com/inspiremenow | inspiremenow.top

Education

EURECOM — Sophia Antipolis, France
Diplôme d'Ingénieur in Digital Sciences — *Sep 2026 — Present*

Shijiazhuang Tiedao University — Hebei, China
B.Eng. in Computer Science and Technology — *Sep 2020 — Jun 2024*

- Weighted Average Score: 81.42/100
- A/A+ Courses: Object-Oriented Programming, Principle and Application of Embedded System, etc.

Skills

- **Proficient in**: C/C++, Python, Shell, SQL, Git/CI/CD
- **Familiar with**: Java, Go, Vue.js, JavaScript, TypeScript, Kotlin, QEMU
- **Knowledge of**: OpenCV(applied), Assembly, Rust

Open Source Contributions

NCNN — Aug 2024 — Dec 2025

- Implemented support for the NCNN_ISA environment variable to enhance hardware compatibility.
- Developed NCNN C API bindings for MicroPython to enable Python-level integration.
- Introduced RenderDoc integration for GPU debugging and profiling support.

Work Experience

QEMU RISC-V Development Intern — Mar 2024 — May 2024
PLCT Lab — *Remote*

- Implemented support for the new rv64ilp32 target in QEMU, which involved porting the existing riscv64 ELF parser to enable compatibility with 32-bit applications.
- Expanded GCC regression testing support to include the rv64ilp32 user mode, reusing the existing riscv64 test suite to perform functional validation.

Related Projects

Maintainer, Video Surveillance System for Study Rooms — Mar 2024 — May 2024

- Architected and developed a client-server surveillance system with server-side YOLO-based real-time person detection, streaming video to Android and desktop clients for monitoring study room occupancy.
- Developed a scalable Go/Gin backend capable of reliably handling concurrent video streams while maintaining a highly responsive API.
- Developed responsive, cross-platform Android (Kotlin) and Electron desktop (JavaScript) clients.

Maintainer, Embedded System Monitoring Software — Jun 2023 — Jul 2023

- Developed a C++/Qt4 real-time monitoring solution for ARM-based embedded devices to visualize key system performance metrics such as CPU load and memory usage.
- Implemented monitoring for key system resources (CPU, memory, disk and processes), providing near real-time data updates while maintaining low CPU overhead.
- The intuitive GUI with real-time charts contributed to a reduction in critical system overload events during testing on the FriendlyARM Tiny4412 development board.

Contributor, Computer Parts Inventory Management System — Dec 2022 — Jan 2023

- Designed and implemented a full-stack inventory management system to streamline product cataloging.
- Developed a responsive Vue.js frontend with dynamic filtering and batch operations, significantly reducing manual data management time.
- Built a robust Java SpringBoot backend with MyBatis, leveraging connection pooling to enhance database query performance and reduce search response times.

- Implemented a secure, JWT-based authentication system with role-based authorization.

Maintainer, Weibo Data Visualization Analysis — Nov 2022 — Dec 2022

- Developed data visualizations with Matplotlib and Pyecharts to analyze Weibo user trends.
- Extracted and visualized high-frequency keywords as word clouds using WordCloud.

Contributor, NeverMiss Scheduling Reminder Software — Nov 2022 — Dec 2022

- Developed a lightweight Windows desktop application for intelligent task scheduling and notification management.
- Engineered a multi-threaded architecture using System.Threading for non-blocking, concurrent task execution.

Awards

IELTS Overall Band Score: 6.0	Apr 2026
Tencent Outstanding Developer Certificate	Nov 2025
Tencent Open Source Contributor Certificate	Oct 2024
Tencent Outstanding Student Certificate	Oct 2024
15th China Undergraduate Computer Design Competition, Provincial Third Prize	Jun 2023
14th Lanqiao Cup National Software Competition, Provincial Third Prize	Apr 2023